Dylan P. Losey

Contact

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Research Interests

Human-robot interaction, robot learning, and control theory, with applications in personal and assistive robots.

Current Position

Virginia Tech
Department of Mechanical Engineering
Associate Professor — 2025 – Present
Assistant Professor — 2020 – 2025
Director of the Collaborative Robotics Laboratory (Collab)

Education

Stanford University
Postdoctoral Scholar in Computer Science — 2019 – 2020
Advisor: Dorsa Sadigh

Rice University
Ph.D. in Mechanical Engineering — 2018
M.S. in Mechanical Engineering — 2016
Dissertation: *Responding to Physical Human-Robot Interaction*
Advisor: Marcia K. O'Malley

Vanderbilt University
B.E. in Mechanical Engineering — 2014

Honors & Awards

Virginia Tech
John R. Jones III Faculty Fellow — 2026

American Society of Mechanical Engineers
Rising Star of Mechanical Engineering — 2025

National Science Foundation
CAREER Award — 2024

IEEE Transactions on Haptics
Best Application Paper Award — 2024

Virginia Tech
Outstanding New Assistant Professor — 2023

IEEE/RSJ International Conference on Intelligent Robots and Systems
Finalist, Best RoboCup Paper Award — 2021

Conference on Robot Learning
Best Paper Award 2020

Robotics: Science and Systems
Finalist, Best Student Paper Award 2020

ACM/IEEE International Conference on Human-Robot Interaction
Honorable Mention, Best Paper Award 2020

Rice University
Outstanding Ph.D. Thesis 2019

IEEE/ASME Transactions on Mechatronics
Best Paper Award 2017

IEEE Conference on Biomedical Robotics and Biomechatronics
Finalist, Best Student Paper Award 2016

National Science Foundation
Graduate Research Fellowship 2014

Vanderbilt University
Dynamics & Controls Award 2014

Vanderbilt University
Cornelius Vanderbilt Scholarship 2010

Teaching

ME 3534: Controls Engineering I *Last Lectured:*
Instructor, Virginia Tech Spring 2025

ME 4524: Robotics & Automation
ME 4584: Robotics Laboratory *Last Lectured:*
Instructor, Virginia Tech Fall 2026

ME 5704 / ECE 5704: Robotics & Automation *Last Lectured:*
Instructor, Virginia Tech Fall 2025

ME 4824: Introduction to Human-Robot Interaction *Last Lectured:*
Course Developer and Instructor, Virginia Tech Spring 2026

ME 5824 / CS 5844: Algorithmic Human-Robot Interaction *Last Lectured:*
Course Developer and Instructor, Virginia Tech Spring 2026

Interdisciplinary Capstone *Most Recent:*
Advisor, Virginia Tech 2024 – 2025

ME 4015 / ME 4016: Engineering Design and Project *Most Recent:*
Advisor, Virginia Tech 2023 - 2024

Advising & Mentoring

Current Ph.D. Students
Benjamin Christie, Robert Ramirez Sanchez, Yinlong Dai, Mohammad Bararjanian-bahnamiri, Jaewoo Cho

Current M.S. Students
Iain Mischel, Sarah Bradshaw, Brage Gundersen

Past Postdoctoral Scholars
Heramb Nemlekar (Assistant Professor at CSUN)

Past Ph.D. Students
Sagar Parekh, Shahabedin Sagheb (Boeing), Soheil Habibian (Apple), Ananth Jonnavittula (Foundation), Shaunak Mehta (Fujitsu)

Past M.S. Students
Maya Keely (Eldor Corporation), Joshua Hoegerman (Boeing), Ritish Shailly (Amazon)

Selected Outreach

Welcome to Mechanical Engineering at VT (WeMET) 2022 – Present
I founded and sponsor this program at Virginia Tech. WeMET provides an opportunity for Virginia Tech graduate students across different research groups to get to know each other, socialize, discuss the highs and lows of graduate school, and perform professional development activities. WeMET aims to support all members of the Virginia Tech ME community, and is committed to fostering an environment where all feel safe and comfortable discussing their graduate journey.

Women's Preview Weekend 2021 – Present
I give talks and host lab visits for this yearly event. During Women's Preview Weekend prospective students who have been offered admission to Virginia Tech visit the campus and learn about ongoing research.

Engineering Open House 2022 – Present
I give talks and host lab visits for this yearly event. Engineering Open House enables admitted high school students to learn more about engineering at Virginia Tech.

RoboGrinder 2023 – Present
I am the faculty advisor for this Virginia Tech student team. The group of around 50 undergraduates from multiple departments designs, builds, and programs robots for national and international competitions (e.g., RoboMaster North America).

Professional Activities

Senior Associate Editor
ACM Transactions on Human-Robot Interaction (THRI) 2025 – Present

Associate Editor, Journal
ACM Transactions on Human-Robot Interaction (THRI) 2021 – 2025
IEEE Robotics and Automation Letters (RA-L) 2020 – 2023

Associate Editor, Conference
IEEE International Conference on Intelligent Robots and Systems (IROS) 2021, 2022
IEEE International Symposium on Multi-Robot and Multi-Agent Systems (MRS) 2021

Virginia Robotics Symposium
Organizer 2025, 2026

Workshop Organizer
ICRA: *Bridging the Gap between Robot Learning and Human-Robot Interaction* 2026
IROS: *Shared Autonomy and the Sense of Agency* 2025
ICRA: *Towards Collaborative Partners: Design, Shared Control, and Robot Learning* 2024
ICRA: *Communicating Robot Learning across Human-Robot Interaction* 2023
RSS: *Emergent Behaviors in Human-Robot Systems* 2020

Center for Human-Computer Interaction
Member, Virginia Tech CHCI 2021 – Present

External Reviewer
NSF Panels, Journals including The International Journal of Robotics Research, IEEE Transactions on Robotics, IEEE Transactions on Robot Learning, IEEE Robotics and Automation Letters, ACM Transactions on Human-Robot Interaction, Autonomous Robots, and IEEE/ASME Transactions on Mechatronics.
Conferences including ICRA, RSS, CoRL, IROS, and HRI.

Invited Talks

NSF M3X Annual Meeting
National Science Foundation 2026
Closing the Loop between Learning and Communication for Assistive Robot Arms

Purdue University
Edwardson School of Industrial Engineering Seminar 2026
What does it take to learn from humans?

Mitsubishi Electric Research Laboratories (MERL)
External Seminar 2026

NSF FRR-NRI Annual Meeting
National Science Foundation 2025
Unifying Rigid and Soft Grippers for Assistive Eating

Imperial College London
Robotics Seminar 2025
What does it take to learn from humans?

University of Utah
Robotics Seminar 2025

Cornell University
MAE Colloquium Seminar Series 2024

The University of Texas at Austin
Oden Institute Seminar 2024
Robots that Learn to Influence Humans

Rice University
Mechanical Engineering Seminar 2024

University of Tennessee, Knoxville
MABE Distinguished Seminar 2024

The Boeing Company

Boeing Distinguished Researcher and Scholar Seminar 2023

University of Waterloo
Mechanical Engineering Seminar 2023

Vanderbilt University
Mechanical Engineering Seminar 2022

Worcester Polytechnic Institute
Robotics Engineering Colloquium 2022

Commonwealth Cyber Initiative
Integrated Security Seminar 2022

University of California, Berkeley
CITRIS People and Robots Seminar 2022

Purdue University
Robotics Seminar 2021
Interactive, Inclusive, and Revealing Robot Learners

University of Illinois Urbana-Champaign
Human-Centered Autonomy Lab 2021

University of Virginia
ESE Colloquium 2021

University of California, Berkeley
InterACT Lab 2021
Towards Inclusive and Revealing Robot Learners

University of California, Berkeley
Model Predictive Control Lab 2020
Latent Roles and Strategies in Multi-Agent Interaction

IEEE International Conference on Robotics and Automation
Workshop on Interactive Robot Learning 2020
Personalizing Robots through Learned Latent Actions

University of North Carolina
Computer Science Seminar 2020
Personalizing Robots with Mechanics and Learning

University of Washington
Mechanical Engineering Seminar 2020

Boston University
Mechanical Engineering Seminar 2020

Notre Dame
Aerospace and Mechanical Engineering Seminar 2020

Virginia Tech
Mechanical Engineering Seminar 2020

Colorado School of Mines
Mechanical Engineering Seminar 2020

Stanford University
Robotics Seminar 2020

Amazon Research Awards 2019
Controlling Assistive Robots with Learned Latent Actions

Massachusetts Institute of Technology
Computer Science & Artificial Intelligence Lab 2019
Personalizing Robots with Physics and Intelligence

Harvard University
Harvard Biodesign Lab 2019

Boston University
Center for Information & Systems Engineering Seminar 2019

University of Illinois Urbana-Champaign
Mechanical Engineering & Computer Science Seminars 2019
Responding to Physical Human-Robot Interaction

Stanford University
Robotics Seminar 2019

Journal Papers

45. Ye-Ji Mun, Mahsa Golchoubian, Shahabedin Sagheb, Yan Bai, Tianhao Ji, Andrea Bajcsy, Dylan P. Losey, and Katherine Driggs-Campbell, "Belief-aware influence and trust (BAIT): Shaping human belief during repeated human-robot interaction," *IEEE Robotics and Automation Letters*, 2026 (in review).
44. Shaunak A. Mehta and Dylan P. Losey, "Combining performance and passivity in linear control of series elastic actuators," *IEEE Robotics and Automation Letters*, 2026 (in review).
43. Benjamin A. Christie, Sagar Parekh, and Dylan P. Losey, "Fine-tuning robot policies while maintaining user privacy," *IEEE Robotics and Automation Letters*, 2026 (in review).
42. Yinlong Dai, Benjamin A. Christie, Daniel J. Evans, Dylan P. Losey, and Simon Stepputtis, "Language movement primitives: Grounding language models in robot motion," *IEEE Transactions on Robot Learning*, 2026 (in review).
41. Benjamin A. Christie and Dylan P. Losey, "Safe interactions via Monte Carlo linear-quadratic games," *ACM Transactions on Human-Robot Interaction*, 2026 (in review).
40. Sagar Parekh, Preston Culbertson, and Dylan P. Losey, "TransMASK: Masked state representation through learned transformation," *IEEE Transactions on Robot Learning*, 2026 (in review).
39. Robert Ramirez Sanchez, Daniel J. Evans, Dylan P. Losey, and Siddarth Jain, "VOLT: Vision and language trajectory segmentation for faster-than-demonstration policies," *Autonomous Robots*, 2026 (in review).
38. Michael Hagenow, Mario Selvaggio, Xuehui Yu, Yanwei Wang, Yiannis Demiris, Andreea Bobu, Yilun Du, Harold Soh, Dylan P. Losey, and Julie Shah, "Shared

control/autonomy in human-robot interaction: Current trends and the role of generative AI," *The International Journal of Robotics Research*, 2026 (in review).

37. Yinlong Dai, Robert Ramirez Sanchez, Ryan Jeronimus, Shahabedin Sagheb, Cara M. Nunez, Heramb Nemlekar, and Dylan P. Losey, "CIVIL: Causal and intuitive visual imitation learning," *Autonomous Robots*, 2025 (in press).
36. Sagar Parekh, Casey Grothoff, Ryan Wright, Robin White, and Dylan P. Losey, "Safe and transparent robots for human-in-the-loop meat processing," *Scientific Reports*, 2025 (in review).
35. Shahabedin Sagheb and Dylan P. Losey, "Counterfactual behavior cloning: Offline imitation learning from imperfect human demonstrations," *ACM Transactions on Human-Robot Interaction*, 2026 (in press).
34. Shahabedin Sagheb, Sagar Parekh, Ravi Pandya, Ye-Ji Mun, Katherine Driggs-Campbell, Andrea Bajcsy, and Dylan P. Losey, "A unified framework for robots that influence humans over long-term interaction," *The International Journal of Robotics Research*, 2025 (in review).
33. Antonio Alvarez Valdivia, Benjamin A. Christie, Dylan P. Losey, and Laura H. Blumenschein, "A modular haptic display with reconfigurable signals for personalized information transfer," *IEEE Transactions on Haptics*, vol. 19, no. 1, 2026.

32 Maya Keely, Brandon Franco, Casey Grothoff, Rajat Kumar Jenamani, Tapomayukh Bhattacharjee, Dylan P. Losey, and Heramb Nemlekar, "Kiri-Spoon: A kirigami utensil for robot-assisted feeding," *The International Journal of Robotics Research*, vol. 45, no. 11, pp. 1772-1797, 2026. **(Featured on Journal Cover)**

31. Maya Keely, Yeunhee Kim, Shaunak A. Mehta, Joshua Hoegerman, Robert R. Sanchez, Emily Paul, Camryn Mills, Dylan P. Losey, and Michael D. Bartlett, "Combining and decoupling rigid and soft grippers to enhance robotic manipulation," *Soft Robotics*, 2025 (in press).
30. Sagar Parekh, Heramb Nemlekar, and Dylan P. Losey, "Towards balanced behavior cloning from imbalanced datasets," *Autonomous Robots*, vol. 50, no. 9, 2026.
29. Ryan Wright, Sagar Parekh, Dylan P. Losey, and Robin White, "Surveying Processor Perceptions of Automation in Meat Processing," *Applied Food Research*, vol. 5, no. 2, 2025.
28. Shaunak A. Mehta, Heramb Nemlekar, Hari Sumant, and Dylan P. Losey, "L2D2: Robot learning from 2D drawings," *Autonomous Robots*, vol. 49, no. 25, 2025.
27. Heramb Nemlekar, Robert Ramirez Sanchez, and Dylan P. Losey, "PECAN: Personalizing robot behaviors through a learned canonical space," *ACM Transactions on Human-Robot Interaction*, vol. 14, no. 4, pp. 1-33, 2025.
26. Shahabedin Sagheb, Soham Gandhi, and Dylan P. Losey, "Should collaborative robots be transparent?," *International Journal of Social Robotics*, vol. 17, pp. 937-953, 2025.
25. Shaunak A. Mehta, Yusuf Ciftci, Balamurugan Ramachandran, Somil Bansal, and Dylan P. Losey, "Stable-BC: Controlling covariate shift with stable behavior cloning," *IEEE Robotics and Automation Letters*, vol. 10, no. 2, pp. 1952-1959, 2025.
24. Ananth Jonnavittula, Sagar Parekh, and Dylan P. Losey, "VIEW: Visual imitation learning with waypoints," *Autonomous Robots*, vol. 49, no. 5, 2025.

23. Soheil Habibian, Antonio Valdivia, Laura Blumenschein, and Dylan P. Losey, "A survey of communicating robot learning during human-robot interaction," *The International Journal of Robotics Research*, vol. 44, no. 4, pp. 665-698, 2025. **(Featured on Journal Cover)**
22. Benjamin A. Christie and Dylan P. Losey, "LIMIT: Learning interfaces to maximize information transfer," *ACM Transactions on Human-Robot Interaction*, vol. 13, no. 4, pp. 1-26, 2024.
21. Shaunak A. Mehta and Dylan P. Losey, "Unified learning from demonstrations, corrections, and preferences during physical human-robot interaction," *ACM Transactions on Human-Robot Interaction*, vol. 13, no. 3, pp. 1-25, 2024.
20. Ananth Jonnavittula, Shaunak A. Mehta, and Dylan P. Losey, "SARI: Shared autonomy across repeated interaction," *ACM Transactions on Human-Robot Interaction*, vol. 13, no. 2, pp. 1-36, 2024.
19. Ryan Wright, Sagar Parekh, Robin White, and Dylan P. Losey, "Safely and autonomously cutting meat with a collaborative robot arm," *Scientific Reports*, vol 14, pp. 299, 2024.
18. Shaunak A. Mehta, Forrest Meng, Andrea Bajcsy, and Dylan P. Losey, "StROL: Stabilized and robust online learning from humans," *IEEE Robotics and Automation Letters*, vol. 9, no. 3, pp. 2303-2310, 2023.
17. Michael Hagenow, Emmanuel Senft, Nitzan Orr, Robert Radwin, Michael Gleicher, Bilge Mutlu, Dylan P. Losey, and Michael Zinn, "Coordinated multi-robot shared autonomy based on scheduling and demonstrations," *IEEE Robotics and Automation Letters*, vol. 8, no. 12, pp. 8335-8342, 2023.
16. Joshua Hoegerman and Dylan P. Losey, "Reward learning with intractable normalizing functions," *IEEE Robotics and Automation Letters*, vol. 8, no. 11, pp. 7511-7518, 2023.
15. Sagar Parekh, and Dylan P. Losey, "Learning latent representations to co-adapt to humans," *Autonomous Robots*, vol. 47, pp. 771–796, 2023.
14. Antonio Valdivia, Soheil Habibian, Carly Mendenhall, Francesco Fuentes, Ritish Shailly, Dylan P. Losey, and Laura Blumenschein, "Wrapping haptic displays around robot arms to communicate learning," *IEEE Transactions on Haptics*, vol. 16, no. 1, pp. 57-72, 2023. **(Best Application Paper Award)**
13. Soheil Habibian, Ananth Jonnavittula, and Dylan P. Losey, "Here's what I've learned: Asking questions that reveal reward learning," *ACM Transactions on Human-Robot Interaction*, vol. 11, no. 4, pp. 1-28, 2022.
12. Soheil Habibian and Dylan P. Losey, "Encouraging human interaction with robot teams: Legible and fair subtask allocations," *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 6685-6692, 2022.
11. Dylan P. Losey, Andrea Bajcsy, Marcia K. O'Malley, and Anca D. Dragan, "Physical interaction as communication: Learning robot objectives online from human corrections," *The International Journal of Robotics Research*, vol. 41, no. 1, pp. 20-44, 2022.
10. Erdem Bıyık, Dylan P. Losey, Malayandi Palan, Nicholas C. Landolfi, and Dorsa Sadigh, "Learning reward functions from diverse sources of human feedback: Optimally integrating demonstrations and preferences," *The International Journal of Robotics Research*, vol. 41, no. 1, pp. 45-67, 2022.
9. Dylan P. Losey, Hong Jun Jeon, Mengxi Li, Krishnan Srinivasan, Ajay Mandlekar, Animesh Garg, Jeannette Bohg, and Dorsa Sadigh, "Learning latent actions to control assistive robots," *Autonomous Robots*, vol. 46, pp. 115–147, 2022.

8. James F. Mullen Jr, Josh Mosier, Sounak Chakrabarti, Anqi Chen, Tyler White, and Dylan P. Losey, "Communicating inferred goals with passive augmented reality and active haptic feedback," *IEEE Robotics and Automation Letters*, vol. 6, no. 4, pp. 8522-8529, 2021.
7. Dylan P. Losey and Marcia K. O'Malley, "Learning the correct robot trajectory in real-time from physical human interactions," *ACM Transactions on Human-Robot Interaction*, vol. 9, no. 1, pp. 1-19, 2019.
6. Dylan P. Losey, Laura Blumenschein, Janelle Clark, and Marcia K. O'Malley, "Improving short-term retention after robotic training by leveraging fixed-gain controllers," *Journal of Rehabilitation and Assistive Technologies Engineering*, vol. 6, pp. 1-13, 2019.
5. Dylan P. Losey and Marcia K. O'Malley, "Enabling robots to infer how end-users teach and learn through human-robot interaction," *IEEE Robotics and Automation Letters*, vol. 4, no. 2, pp. 1956-1963, 2019.
4. Dylan P. Losey and Marcia K. O'Malley, "Trajectory deformations from physical human-robot interaction," *IEEE Transactions on Robotics*, vol. 34, no. 1, pp. 126-138, 2018.
3. Dylan P. Losey, Craig G. McDonald, Edoardo Battaglia, and Marcia K. O'Malley, "A review of intent detection, arbitration, and communication aspects of shared control for physical human-robot interaction," *Applied Mechanics Reviews*, vol. 70, no. 1, 2018.
2. Dylan P. Losey, Andrew Erwin, Craig G. McDonald, Fabrizio Sergi, and Marcia K. O'Malley, "A time domain approach to control of series elastic actuators: Adaptive torque and passivity-based impedance control," *IEEE/ASME Transactions on Mechatronics*, vol. 21, no. 4, pp. 2085-2096, 2016. **(Best Paper Award)**.
1. Ali Utku Pehlivan, Dylan P. Losey, and Marcia K. O'Malley, "Minimal assist-as-needed controller for upper limb robotic rehabilitation," *IEEE Transactions on Robotics*, vol. 32, no. 1, pp. 113-124, 2016.

Refereed Conference Proceedings

44. Yinlong Dai, Homanga Bharadhwaj, and Dylan P. Losey, "CASTER: Formalizing zero-shot manipulation from synthetic videos as constrained optimization," *IEEE International Conference on Robotics and Automation (ICRA)*, 2026 (under review).
43. Mengxue Fu, Ethan Xu, Sam Iyer-Singh, Yinlong Dai, Michael Hagenow, and Dylan P. Losey, "Assisting for open-ended tasks: Goal-oriented shared autonomy as a particle filter," *IEEE International Conference on Robotics and Automation (ICRA)*, 2026 (under review).
42. Daniel J. Evans, Yinlong Dai, Simon Stepputtis, and Dylan P. Losey, "GraspTwin: Zero-shot task-oriented grasp optimization via a digital twin," *IEEE International Conference on Robotics and Automation (ICRA)*, 2026 (under review).
41. Benjamin A. Christie, Siddarth Jain, and Dylan P. Losey, "Show and tell: Unifying physical demonstrations and language corrections for neuro-symbolic robots," *IEEE International Conference on Robotics and Automation (ICRA)*, 2026 (under review).
40. Sung-Wook Lee, Xuhui Kang, Dylan P. Losey, and Yen-Ling Kuo, "PAIR: Proprioception-aligned image-ray representations for viewpoint-robust manipulation," *IEEE International Conference on Robotics and Automation (ICRA)*, 2026 (under review).

39. Mengxue Fu, Ya-Chuan Hsu, Yinlong Dai, Erdem Bıyık, and Dylan P. Losey, "In view or out of view? A vision-aware filter that infers and reacts to the human's field of view," *IEEE International Conference on Robotics and Automation (ICRA)*, 2026 (under review).
38. Benjamin A. Christie, Yinlong Dai, Mohammad Bararjanianbahnamiri, Simon Stepputtis, and Dylan P. Losey, "From local corrections to generalized skills: Improving neuro-symbolic policies with MEMO," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
37. Keone Leao, Grace Brotherson, Iain Mischel, Sagar Parekh, and Dylan P. Losey, "A pivot-based kirigami utensil for hand-held and robot-assisted feeding," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
36. Luoyan Zhong, Heather Jin Hee Kim, Dylan P. Losey, and Cara M. Nunez, "Curvature-aware calibration of tactile sensors for accurate force estimation on non-planar surfaces," *Haptics Symposium*, 2026.
35. Yinlong Dai, Andre Keyser, and Dylan P. Losey, "Prepare before you act: Learning from humans to rearrange initial states," *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
34. Sagar Parekh, Lauren Bramblett, Nicola Bezzo, and Dylan P. Losey, "Using high-level patterns to estimate how humans predict a robot will behave," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
33. Robert Ramirez Sanchez, Heramb Nemlekar, Shahabedin Sagheb, and Dylan P. Losey, "RECON: Reducing causal confusion with human-placed markers," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
32. Benjamin A. Christie, Heramb Nemlekar, and Dylan P. Losey, "Personalizing interfaces to humans with user-friendly priors," *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
31. Joshua Hoegerman, Shahabedin Sagheb, Benjamin A. Christie, and Dylan P. Losey, "Aligning learning with communication in shared autonomy," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.
30. Shaunak A. Mehta, Soheil Habibian, and Dylan P. Losey, "Waypoint-based reinforcement learning for robot manipulation tasks," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.
29. Maya N. Keely, Heramb Nemlekar, and Dylan P. Losey, "Kiri-Spoon: A soft shape-changing utensil for robot-assisted feeding," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.
28. Shaunak A. Mehta, Yeunhee Kim, Joshua Hoegerman, Michael D. Bartlett, and Dylan P. Losey, "RISO: Combining rigid grippers with soft switchable adhesives," *IEEE International Conference on Soft Robotics (RoboSoft)*, 2023.
27. Shahabedin Sagheb, Ye-Ji Mun, Neema Ahmadian, Benjamin A. Christie, Andrea Bajcsy, Katherine Driggs-Campbell, and Dylan P. Losey, "Towards robots that influence humans over long-term interaction," *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
26. Sagar Parekh, Soheil Habibian, and Dylan P. Losey, "RILI: Robustly influencing latent intent," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.
25. Antonio Alvarez Valdivia, Ritish Shailly, Naman Seth, Francesco Fuentes, Dylan P. Losey, and Laura H. Blumenschein, "Wrapped haptic display for communicating physical robot learning," *IEEE/RAS International Conference on Soft Robotics (RoboSoft)*, 2022.

24. Ehsan Yousefi, Dylan P. Losey, and Inna Sharf, "Assisting operators of articulated machinery with optimal planning and goal inference," *IEEE International Conference on Robotics and Automation (ICRA)*, 2022.
23. Shaunak A. Mehta, Sagar Parekh, and Dylan P. Losey, "Learning latent actions without human demonstrations," *IEEE International Conference on Robotics and Automation (ICRA)*, 2022.
22. Ananth Jonnavittula and Dylan P. Losey, "Communicating robot conventions through shared autonomy," *IEEE International Conference on Robotics and Automation (ICRA)*, 2022.
21. Ananth Jonnavittula and Dylan P. Losey, "Learning to share autonomy across repeated interaction," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021. **(Finalist, Best RoboCup Paper)**
20. Siddharth Karamcheti, Albert J. Zhai, Dylan P. Losey, and Dorsa Sadigh, "Learning visually guided latent actions for assistive teleoperation," *Learning for Dynamics & Control (L4DC)*, 2021.
19. Ananth Jonnavittula and Dylan P. Losey, "I know what you meant: Learning human objectives by (under)estimating their choice set," *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.
18. Mengxi Li, Alper Canberk, Dylan P. Losey, and Dorsa Sadigh, "Learning human objectives from sequences of physical corrections," *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.
17. Annie Xie, Dylan P. Losey, Ryan Tolsma, Chelsea Finn, and Dorsa Sadigh, "Learning Latent Representations to Influence Multi-Agent Interaction," *Conference on Robot Learning (CoRL)*, 2020. **(Best Paper Award)**
16. Mengxi Li, Dylan P. Losey, Jeannette Bohg, and Dorsa Sadigh, "Learning user-preferred mappings for intuitive robot control," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020.
15. Hong Jun Jeon, Dylan P. Losey, and Dorsa Sadigh, "Shared autonomy with learned latent actions," *Robotics: Science and Systems (RSS)*, 2020. **(Finalist, Best Student Paper)**
14. Dylan P. Losey, Krishnan Srinivasan, Ajay Mandlekar, Animesh Garg, and Dorsa Sadigh, "Controlling assistive robots with learned latent actions," *IEEE International Conference on Robotics and Automation (ICRA)*, 2020.
13. Minae Kwon, Erdem Bıyık, Aditi Talati, Karan Bhasin, Dylan P. Losey, and Dorsa Sadigh, "When humans aren't optimal: Robots that collaborate with risk-aware humans," *ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2020. **(Honorable Mention, Best Paper)**
12. Dylan P. Losey, Mengxi Li, Jeannette Bohg, and Dorsa Sadigh, "Learning from my partner's actions: Roles in decentralized robot teams," *Conference on Robot Learning (CoRL)*, 2019.
11. Erdem Bıyık, Malayandi Palan, Nicholas C. Landolfi, Dylan P. Losey, and Dorsa Sadigh, "Asking easy questions: A user-friendly approach to active reward learning," *Conference on Robot Learning (CoRL)*, 2019.
10. Dylan P. Losey and Dorsa Sadigh, "Robots that take advantage of human trust," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2019.
9. Dylan P. Losey and Marcia K. O'Malley, "Including uncertainty when learning from human corrections," *Conference on Robot Learning (CoRL)*, 2018.

8. Andrea Bajcsy, Dylan P. Losey, Marcia K. O'Malley, and Anca D. Dragan, "Learning from physical human corrections, one feature at a time," *ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2018.
7. Andrea Bajcsy, Dylan P. Losey, Marcia K. O'Malley, and Anca D. Dragan, "Learning robot objectives from physical human interaction," *Conference on Robot Learning (CoRL)*, 2017.
6. Dylan P. Losey and Marcia K. O'Malley, "Effects of discretization on the K-width of series elastic actuators," *IEEE International Conference on Robotics and Automation (ICRA)*, 2017.
5. Ali Utku Pehlivan, Dylan P. Losey, Chad G. Rose, and Marcia K. O'Malley, "Maintaining subject engagement during robotic rehabilitation with a minimal assist-as-needed (mAAN) controller," *IEEE International Conference on Rehabilitation Robotics (ICORR)*, 2017.
4. Dylan P. Losey, Laura H. Blumenschein, and Marcia K. O'Malley, "Improving the retention of motor skills after reward-based reinforcement by incorporating haptic guidance and error augmentation," *IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob)*, 2016.
3. Dylan P. Losey, Craig G. McDonald, and Marcia K. O'Malley, "A bio-inspired algorithm for identifying unknown kinematics from a discrete set of candidate models by using collision detection," *IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob)*, 2016. **(Finalist, Best Student Paper Award)**
2. Ben D. Kramer, Dylan P. Losey, and Marcia K. O'Malley, "SOM and LVQ classification of endovascular surgeons using motion-based metrics," *Workshop on Self-Organizing Maps (WSOM)*, 2016.
1. Dylan P. Losey, Peter A. York, Philip J. Swaney, Jessica Burgner, and Robert J. Webster III, "A flexure-based wrist for needle-sized surgical robots," *SPIE Medical Imaging*, 2013.